

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 41

When does formation of primary spermatocytes begin?

- A. During week 4 of embryonic life
- B. During month 5 of infancy
- C. At puberty
- D. During month 5 of fetal life
- E. At birth

Correct Answer: C. At puberty

Explanation: In males, primordial germ cells remain dormant in the seminiferous tubules of the testes from the fetal period until puberty. At **puberty**, these cells differentiate into spermatogonia, which then undergo mitotic divisions and begin differentiating into **primary spermatocytes** to initiate spermatogenesis.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 42

When does a secondary oocyte complete its second meiotic division to become a mature ovum?

- A. At fertilization
- B. At puberty
- C. Before birth
- D. At ovulation
- E. Before ovulation

Correct Answer: A. At fertilization

Explanation:

◇ During oogenesis, a primary oocyte completes its first meiotic division just before ovulation to become a **secondary oocyte** (arrested in metaphase II).

◇ The secondary oocyte will only complete its second meiotic division to become a mature ovum **at the moment of fertilization**, when a sperm successfully penetrates its cell membrane. If fertilization does not occur, the cell degenerates without completing meiosis II.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 43

Which of the following events is involved in cleavage of the zygote during Week 1 of development?

- A. A series of meiotic divisions forming blastomeres
- B. An increased cytoplasmic content of blastomeres
- C. An increase in size of blastomeres
- D. A decrease in size of blastomeres

Correct Answer: D. A decrease in size of blastomeres

Explanation:

✓ **Cleavage** consists of a series of rapid **mitotic** (not meiotic) divisions of the zygote within the rigid, non-expanding boundaries of the *zona pellucida*.

✓ Because there is no cell growth between these rapid divisions, the cytoplasm is partitioned into increasingly smaller cells. Therefore, with each successive cleavage division, there is a **progressive decrease in the size of the individual daughter cells (blastomeres)**.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 44

The first indication of gastrulation in the embryo is

- A. Formation of the primitive streak
- B. Formation of the neural tube
- C. Formation of extraembryonic mesoderm
- D. Formation of tertiary chorionic villi
- E. Formation of the notochord

Correct Answer: A. Formation of the primitive streak

Explanation: Gastrulation is the landmark process during Week 3 of development where the bilaminar embryonic disc is converted into a trilaminar embryonic disc. The very first morphological sign of this process is the **formation of the primitive streak** on the dorsal surface of the epiblast.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 45

The epiblast is capable of forming which of the following germ layers?

- A. Ectoderm only
- B. Ectoderm and mesoderm only
- C. Ectoderm, mesoderm, endoderm
- D. Mesoderm and endoderm only
- E. Ectoderm and endoderm only

Correct Answer: C. Ectoderm, mesoderm, endoderm

Explanation: The **epiblast** is the ultimate source of all three germ layers in the trilaminar embryo. During gastrulation, cells of the epiblast migrate downward through the primitive

streak; some displace the hypoblast to form the **embryonic endoderm**, others interpose in the middle to form the **mesoderm**, and the remaining cells left on the surface become the **ectoderm**.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 46

In the production of male gametes, which of the following cells remains dormant for 14 years?

- A. Primary spermatocyte
- B. Primordial germ cell
- C. Secondary spermatocyte
- D. Spermatid
- E. Sperm

Correct Answer: B. Primordial germ cell

Explanation:

✓ In males, the **primordial germ cells** migrate from the yolk sac to the genital ridges during the 4th to 6th weeks of embryonic life. Once inside the primitive seminiferous tubules of the testes, they remain completely **dormant** for about **14 years** until **puberty**, when they begin differentiating into spermatogonia to launch active spermatogenesis.

✓ *Primary spermatocytes* (Option A) only begin forming at puberty and do not remain dormant.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 47

Which structure is derived from the same embryonic primordium as the dorsal root ganglia?

- A. Kidney
- B. Gonads
- C. Pineal gland
- D. Liver
- E. Adrenal medulla

Correct Answer: E. Adrenal medulla

Explanation:

✓ Adrenal Medulla: Both the dorsal root ganglia and the chromaffin cells of the adrenal medulla are derived from neural crest cells (specifically trunk neural crest cells)

✓ Kidney & Gonads: Both of these structures develop from the intermediate mesoderm (specifically from the urogenital ridge)

✓ Pineal Gland: This gland is a neuroectoderm derivative that develops from the diencephalon (epithalamus)

✓ Liver: The parenchyma of the liver arises from endoderm as an outgrowth of the foregut (the hepatic diverticulum)

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 48

External genitalia organs are developed during-----

- A. First month of pregnancy
- B. First trimester
- C. Second trimester
- D. Third trimester

Correct Answer: B. First trimester

Explanation:

✓ The development of the external genitalia begins early in embryonic life. while male and female embryos are phenotypically indistinguishable between weeks 1 and 6, "**male and female characteristics of the external genitalia can be recognized by week 12**".

✓ Because week 12 marks the end of the **first trimester** (weeks 1–12), the correct answer is the first trimester.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 49

Very often the first indication a woman has that she is pregnant is a missed menstrual period. In which week of embryonic development will a woman experience her first missed menstrual period?

- A. Start of week 3
- B. Start of week 4
- C. Start of week 5
- D. Start of week 8
- E. End of week 8

Correct Answer: A. Start of week 3

Explanation:

✓ A typical menstrual cycle lasts 28 days, with ovulation and fertilization occurring around day 14.

✓ A woman's next expected menstrual period would normally begin 14 days after ovulation (on day 28 of her cycle). This day corresponds exactly to 14 days of embryonic development, which marks the end of embryonic week 2 and the **start of week 3 of embryonic development**.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 50

During the later stages of pregnancy, maternal blood is separated from fetal blood by the-----

- A. Syncytiotrophoblast only
- B. Cytotrophoblast only
- C. Syncytiotrophoblast and cytotrophoblast
- D. Syncytiotrophoblast and fetal endothelium
- E. Cytotrophoblast and fetal endothelium

Correct Answer: D. Syncytiotrophoblast and fetal endothelium

Explanation:

✓ Early in pregnancy, the placental membrane (barrier) consists of four layers: syncytiotrophoblast, cytotrophoblast, extraembryonic mesodermal connective tissue, and fetal capillary endothelium.

✓ In the **later stages of pregnancy**, the cytotrophoblast degenerates and the connective tissue layer is significantly reduced. This leaves maternal and fetal blood separated primarily by only two thin layers: the **syncytiotrophoblast** and the **fetal capillary endothelium**, facilitating more efficient nutrient and gas exchange.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 51

A 30-year-old pregnant woman experiences repeated episodes of bright red vaginal bleeding at week 26, week 31, and week 32 of pregnancy. The bleeding spontaneously subsided each time. Use of ultrasound shows that the placenta is located in the lower right portion of the uterus over the internal os. What is the diagnosis?

- A. Hydatidiform mole
- B. Vasa previa
- C. Placenta previa
- D. Placental abruption
- E. Premature rupture of the amniochorionic membrane

Correct Answer: C. Placenta previa

Explanation: Placenta previa is defined as the implantation of the placenta in the lower uterine segment, where it partially or completely covers the **internal cervical os**. Classically, it presents as repeated, painless episodes of bright red vaginal bleeding in the late second or third trimester (such as weeks 26, 31, and 32) that spontaneously subside.

◇ *Vasa previa* (Option B) occurs when fetal blood vessels cross the internal os.

◇ *Placental abruption* (Option D) is the premature separation of a normally implanted placenta and is typically accompanied by severe abdominal pain and uterine rigidity.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 52

Which of the following time intervals best describes the maximum susceptibility period?

- A. Week 1
- B. Weeks 3–8
- C. Weeks 2–38
- D. Weeks 9–birth

Correct Answer: B. Weeks 3–8

Explanation: The period of **maximum susceptibility to teratogens** (agents that cause congenital malformations) is the **embryonic period**, which spans **weeks 3 to 8** of development. This is the critical window of **organogenesis**, during which all the major organ systems are established.

◇ *Week 1* (Option A) is the pre-embryonic cleavage and blastocyst stage, which operates on an "all-or-none" principle (exposure either kills the blastocyst or the cells recover completely).

◇ *Weeks 9 to birth* (Option D) represent the fetal period, where teratogenic exposure generally results in functional deficiencies or minor growth delays rather than major structural anomalies.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 53

Which of the following structures produces progesterone late in pregnancy?

- A. Placenta
- B. Corpus luteum
- C. Syncytiotrophoblast
- D. Fetal adenohypophysis
- E. Maternal liver

Correct Answer: A. Placenta (specifically via its **C. Syncytiotrophoblast** layer)

Explanation: During the first 6 to 8 weeks of pregnancy, the **corpus luteum** of the ovary (Option B) produces the progesterone required to maintain the pregnancy.

✓ However, late in pregnancy (by the second and third trimesters), the **placenta** takes over this job entirely. If you are looking at tissue-level cellular structures, the **syncytiotrophoblast**

is the specific cell layer of the placenta that synthesizes and secretes this progesterone, but at the organ/structure level, the **placenta** is the correct answer.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 54

Human chorionic gonadotropin (hCG) is produced by which of the following?

- A. Ectoderm
- B. Cytotrophoblast
- C. Decidua basalis
- D. Syncytiotrophoblast
- E. Mesoderm

Correct Answer: D. Syncytiotrophoblast

Explanation: Human chorionic gonadotropin (hCG) is produced and secreted by the **syncytiotrophoblast** cells of the developing placenta. hCG mimics luteinizing hormone (LH) to rescue and maintain the corpus luteum of the ovary so that it continues to secrete progesterone early in pregnancy before the placenta can take over. This is the hormone detected in maternal urine and blood to confirm pregnancy.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 55

Which of the following is the origin of the mitochondrial DNA of all human adult cells?

- A. Paternal only
- B. Maternal only
- C. A combination of paternal and maternal
- D. Either paternal or maternal
- E. Unknown origin

Correct Answer: B. Maternal only

Explanation: Mitochondrial DNA (mtDNA) is inherited exclusively from the mother (**maternal inheritance**). During fertilization, the sperm contributes only its nuclear DNA; any paternal mitochondria located in the sperm's tail/midpiece are either left outside the oocyte during penetration or are selectively destroyed inside the cytoplasm of the fertilized egg. Therefore, all of the mitochondria in an adult's body are clonal descendants of those originally present in the mother's oocyte.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 56

When does formation of primary spermatocytes begin?

- A. During week 4 of embryonic life
- B. During month 5 of fetal life

- C. At birth
- D. During month 5 of infancy
- E. At puberty

Correct Answer: E. At puberty

Explanation: In males, primordial germ cells remain dormant within the seminiferous tubules of the testes from the fetal period until **puberty**. Only at puberty does spermatogenesis actively begin; the dormant cells differentiate into spermatogonia, which then undergo mitotic divisions to give rise to **primary spermatocytes**.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 57

When does a secondary oocyte complete its second meiotic division to become a mature ovum?

- A. At ovulation
- B. Before ovulation
- C. At fertilization
- D. At puberty
- E. Before birth

Correct Answer: C. At fertilization

Explanation: At ovulation, the secondary oocyte is arrested in the metaphase of meiosis II. It will only complete its second meiotic division to become a mature ovum (and release the second polar body) upon **fertilization** when a sperm successfully penetrates its cell membrane. If fertilization does not occur, the secondary oocyte degenerates.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 58

The prochordal plate marks the site of the future

- A. umbilical cord
- B. heart
- C. mouth
- D. anus
- E. nose

Correct Answer: C. mouth

Explanation: The **prochordal (or prechordal) plate** is a thickened organizer area of endoderm at the cranial end of the bilaminar embryonic disc where the ectoderm and endoderm are firmly attached. It marks the future site of the **buccopharyngeal (oropharyngeal) membrane**, which eventually ruptures to form the **mouth**.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 59

Which of the following components plays the most active role in invading the endometrium during blastocyst implantation?

- A. Epiblast
- B. Syncytiotrophoblast
- C. Hypoblast
- D. Extraembryonic somatic mesoderm
- E. Extraembryonic visceral mesoderm

Correct Answer: B. Syncytiotrophoblast

Explanation: During implantation, the outer cell layer of the blastocyst (the trophoblast) differentiates into an inner cytotrophoblast and an outer **syncytiotrophoblast**. The syncytiotrophoblast is a multinucleated, invasive protoplasmic mass that secretes proteolytic enzymes to actively erode and invade the maternal endometrium, allowing the blastocyst to embed.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 60

Which germ layers are present at the end of week 3 of development (day 21)?

- A. Epiblast only
- B. Epiblast and hypoblast
- C. Ectoderm and endoderm
- D. Ectoderm, mesoderm, and endoderm
- E. Epiblast, mesoderm, and hypoblast

Correct Answer: D. Ectoderm, mesoderm, and endoderm

Explanation: Week 3 of development is characterized by gastrulation, the landmark process that converts the bilaminar embryonic disc into a trilaminar embryonic disc. By the end of Week 3 (day 21), all three primary germ layers—**ectoderm, mesoderm, and endoderm**—are fully established and present in the embryo.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 61

Which process establishes the three definitive germ layers?

- A. Neurulation

- B. Gastrulation
- C. Craniocaudal folding
- D. Lateral folding
- E. Angiogenesis

Correct Answer: B. Gastrulation

Explanation: **Gastrulation** is the landmark process occurring during Week 3 of development that converts the bilaminar embryonic disc into a trilaminar embryonic disc, thereby establishing the three definitive germ layers: **ectoderm, mesoderm, and endoderm**.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 62

Intermediate mesoderm give rise to the

- A. neural tube
- B. kidneys and gonads
- C. heart
- D. somites
- E. notochord

Correct Answer: B. kidneys and gonads

Explanation: The embryonic mesoderm differentiates into paraxial, intermediate, and lateral plate mesoderm. The **intermediate mesoderm** specifically runs longitudinally as the nephrogenic cord and gives rise to the **kidneys and gonads**.

[24/09/2026 10:09 AM] Imaanaa Harmee: Question 63

The first indication of gastrulation in the embryo is

- A. formation of the primitive streak
- B. formation of the notochord
- C. formation of the neural tube
- D. formation of extraembryonic mesoderm
- E. formation of tertiary chorionic villi

Correct Answer: A. formation of the primitive streak

Explanation: The very first morphological sign and indication of gastrulation in the third week of development is the **formation of the primitive streak** on the dorsal surface of the epiblast.

[24/09/2026 10:14 AM] Imaanaa Harmee: Question 64

The intervillous space of the placenta contains

- A. maternal blood
- B. fetal blood
- C. maternal and fetal blood
- D. amniotic fluid
- E. maternal blood and amniotic fluid

Correct Answer: A. maternal blood

Explanation: The **intervillous space** of the placenta is filled entirely with **maternal blood** entering from the spiral endometrial arteries. This maternal blood directly bathes the branches of the fetal chorionic villi to facilitate exchange, while fetal blood remains completely enclosed within the capillaries of those villi.

[24/09/2026 10:14 AM] Imaanaa Harmee: Question 65

Which of the following regarding developmental anomalies is NOT true?

- A. Dysplasia is defects caused by abnormal organization of cells into abnormal tissue(s)
- B. Malformation occurring during formation of structures
- C. Disruptions defects resulting in morphological alterations of structure after their formation
- D. Deformations is due to mechanical forces that mold a part of the fetus over a prolonged period of time
- E. None of the above

Correct Answer: E. None of the above

Explanation: All of the provided definitions of developmental anomalies are highly accurate:

- ◇ **Dysplasia** involves the abnormal organization of cells into tissues (abnormal histogenesis).
 - ◇ **Malformations** are structural defects resulting from intrinsically abnormal developmental processes during organ formation.
 - ◇ **Disruptions** are morphological alterations to structures that occurred after their initial normal formation (due to extrinsic interference).
 - ◇ **Deformations** are structural alterations caused by prolonged mechanical molding forces (like uterine compression).
- Since all options (A through D) are true, **None of the above** is the correct choice.

[24/09/2026 10:14 AM] Imaanaa Harmee: Question 66

Which of the following statement about the principles of teratology is NOT true?

- A. Susceptibility to teratogenesis depends on the genotype of the conceptus
- B. The maternal genome plays an important role with respect to drug metabolism, resistance to infection, and other processes that affect the conceptus
- C. Susceptibility to teratogenesis depends on the development stage at the time of exposure
- D. Manifestation of abnormal development depends on duration of exposure and not dose of exposure to a teratogen
- E. Manifestations of abnormal development include death, malformation, growth retardation, and function disorders

Correct Answer: D. Manifestation of abnormal development depends on duration of exposure and not dose of exposure to a teratogen

Explanation :

According to the Principles of Teratology , manifestations of abnormal development depend on both the dose and the duration of exposure to a teratogen

◇ Therefore, the statement in option D is false because it incorrectly claims that the dose does not play a role.

Why the Other Statements Are True:

◇ A and B are TRUE: Susceptibility to teratogenesis depends on the genotype of the conceptus and its interaction with the environment

✓ Additionally, the maternal genome is critically important for drug metabolism, resisting infections, and other biochemical/molecular processes that directly affect the conceptus

◇ C is TRUE: Susceptibility changes based on the developmental stage of the embryo or fetus at the time of exposure

✓ For instance, the first week is a "resistant" period following an "all-or-none" pattern

✓ weeks 3 to 8 represent the period of maximum susceptibility (organogenesis)

✓ and weeks 9 to 38 represent a period of lowered susceptibility

◇ E is TRUE: The manifestations of abnormal development as death, malformation, growth retardation, and functional disorders

[24/09/2026 10:14 AM] Imaanaa Harmee: Question 67

The incidence of major congenital anomalies of major organs are relatively higher in:

A. Heart

- B. Kidneys
- C. Brain
- D. None of the above

Correct Answer: A. Heart

Explanation: Statistically, **congenital heart defects (CHDs)** represent the most common category of major single-organ congenital anomalies, occurring in approximately 1 in 100 (1%) live births globally.

[24/09/2026 10:14 AM] Imaanaa Harmee: Question 68

Which of the following is not true about birth defects?

- A. Short stature is one of the classic features of the Turner syndrome.
- B. A weak catlike cry and microcephaly indicate the clinical diagnosis of Achondroplasia.
- C. Sparse body hair and gynecomastia may indicate the clinical diagnosis of Klinefelter's syndrome.
- D. Tetracycline would cause stained teeth and hypoplasia of enamel.
- E. Microtia is a minor anomaly

Correct Answer: B. A weak catlike cry and microcephaly indicate the clinical diagnosis of Achondroplasia.

Explanation: This statement is **false** (making it the correct choice).

◇ A weak, high-pitched **catlike cry (cri du chat)** along with microcephaly are classic diagnostic signs of **Cri-du-chat syndrome** (caused by a deletion on chromosome 5p).

◇ **Achondroplasia** is a genetic bone-growth disorder resulting in short-limb dwarfism and macrocephaly (large head), not microcephaly or a catlike cry.

◇ All other options are true: short stature is a hallmark of Turner syndrome; gynecomastia and sparse body hair characterize **Klinefelter's syndrome**; tetracycline during development causes teeth staining and enamel hypoplasia; and microtia is classified as a minor anomaly.